

INFORMATION LITERACY



Gess
Community College of Allegheny County

Book: Information Literacy (Gess)

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Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: Chapter 1

1.1: What is Information Literacy?

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CHAPTER OVERVIEW

1.1: What is Information Literacy?

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CHAPTER OVERVIEW

2: Chapter 2

[2.1: Strategies for Gathering Reliable Information](#)

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2.1: Strategies for Gathering Reliable Information

Learning Objectives

1. Distinguish between primary and secondary sources.
2. Identify strategies for locating relevant print and electronic resources efficiently.
3. Identify instances when it is appropriate to use human sources, such as interviews or eyewitness testimony.
4. Identify criteria for evaluating research resources.
5. Understand why many electronic resources are not reliable.

Now that you have planned your research project, you are ready to begin the research. This phase can be both exciting and challenging. As you read this section, you will learn ways to locate sources efficiently, so you have enough time to read the sources, take notes, and think about how to use the information.

Of course, the technological advances of the past few decades—particularly the rise of online media—mean that, as a twenty-first-century student, you have countless sources of information available at your fingertips. But how can you tell whether a source is reliable? This section will discuss strategies for evaluating sources critically so that you can be a media-savvy researcher.

In this section, you will locate and evaluate resources for your paper and begin taking notes. As you read, begin gathering print and electronic resources, identify at least eight to ten sources by the time you finish the chapter, and begin taking notes on your research findings.

Locating Useful Resources

When you chose a paper topic and determined your research questions, you conducted preliminary research to stimulate your thinking. Your research proposal included some general ideas for how to go about your research—for instance, interviewing an expert in the field or analyzing the content of popular magazines. You may even have identified a few potential sources. Now it is time to conduct a more focused, systematic search for informative primary and secondary sources.

Using Primary and Secondary Sources

Writers classify research resources in two categories: primary sources and secondary sources. *Primary sources* are direct, firsthand sources of information or data. For example, if you were writing a paper about the First Amendment right to freedom of speech, the text of the First Amendment in the Bill of Rights would be a primary source.

Other primary sources include the following:

- Research articles
- Literary texts
- Historical documents such as diaries or letters
- Autobiographies or other personal accounts

Secondary sources discuss, interpret, analyze, consolidate, or otherwise rework information from primary sources. In researching a paper about the First Amendment, you might read articles about legal cases that involved First Amendment rights, or editorials expressing commentary on the First Amendment. These sources would be considered secondary sources because they are one step removed from the primary source of information.

The following are examples of secondary sources:

- Magazine articles
- Biographical books
- Literary and scientific reviews
- Television documentaries

Your topic and purpose determine whether you must cite both primary and secondary sources in your paper. Ask yourself which sources are most likely to provide the information that will answer your research questions. If you are writing a research paper about reality television shows, you will need to use some reality shows as a primary source, but secondary sources, such as a reviewer's critique, are also important. If you are writing about the health effects of nicotine, you will probably want to read the published results of scientific studies, but secondary sources, such as magazine articles discussing the outcome of a recent study, may also be helpful.

Once you have thought about what kinds of sources are most likely to help you answer your research questions, you may begin your search for print and electronic resources. The challenge here is to conduct your search efficiently. Writers use strategies to help them find the sources that are most relevant and reliable while steering clear of sources that will not be useful.

Finding Print Resources

Print resources include a vast array of documents and publications. Regardless of your topic, you will consult some print resources as part of your research. (You will use electronic sources as well, but it is not wise to limit yourself to electronic sources only, because some potentially useful sources may be available only in print form.) Table 11.1 lists different types of print resources available at public and university libraries.

Table 11.1 Library Print Resources

Resource Type	Description	Example(s)
Reference works	Reference works provide a summary of information about a particular topic. Almanacs, encyclopedias, atlases, medical reference books, and scientific abstracts are examples of reference works. In some cases, reference books may not be checked out of a library. Note that reference works are many steps removed from original primary sources and are often brief, so these should be used only as a starting point when you gather information.	• <i>The World Almanac and Book of Facts 2010</i> • <i>Diagnostic and Statistical Manual</i> published by the American Psychiatric Association
Nonfiction books	Nonfiction books provide in-depth coverage of a topic. Trade books, biographies, and how-to guides are usually written for a general audience. Scholarly books and scientific studies are usually written for an audience that has specialized knowledge of a topic.	• <i>The Low-Carb Solution: A Slimmer You in 30 Days</i> • <i>Carbohydrates, Fats and Proteins: Exploring the Relationship Between Macronutrient Ratios and Health Outcomes</i>
Periodicals and news sources	These sources are published at regular intervals—daily, weekly, monthly, or quarterly. Newspapers, magazines, and academic journals are examples. Some periodicals provide articles on subjects of general interest, while others are more specialized.	• <i>New York Times</i> • <i>PC Magazine</i> • <i>JAMA, The Journal of the American Medical Association</i>
Government publications	Federal, state, and local government agencies publish information on a variety of topics. Government publications include reports, legislation, court documents, public records, statistics, studies, guides, programs, and forms.	• <i>The Census 2000 Profile</i> • <i>The Business Relocation Package</i> published by the Philadelphia Chamber of Commerce

Resource Type	Description	Example(s)
Business and nonprofit publications	Businesses and nonprofit organizations produce publications designed to market a product, provide background about the organization, provide information on topics connected to the organization, or promote a cause. These publications include reports, newsletters, advertisements, manuals, brochures, and other print documents.	- A company's instruction manual explaining how to use a specific software program - A news release published by the Sierra Club

Some of these resources are also widely available in electronic format. In addition to the resources noted in the table, library holdings may include primary texts such as historical documents, letters, and diaries.

Writing at Work

Businesses, government organizations, and nonprofit organizations produce published materials that range from brief advertisements and brochures to lengthy, detailed reports. In many cases, producing these publications requires research. A corporation's annual report may include research about economic or industry trends. A charitable organization may use information from research in materials sent to potential donors.

Regardless of the industry you work in, you may be asked to assist in developing materials for publication. Often, incorporating research in these documents can make them more effective in informing or persuading readers.

Tip

As you gather information, strive for a balance of accessible, easy-to-read sources and more specialized, challenging sources. Relying solely on lightweight books and articles written for a general audience will drastically limit the range of useful, substantial information. On the other hand, restricting oneself to dense, scholarly works could make the process of researching extremely time-consuming and frustrating.

Exercise 1:

Make a list of five types of print resources you could use to find information about your research topic. Include at least one primary source. Be as specific as possible—if you have a particular resource or type of resource in mind, describe it.

To find print resources efficiently, first identify the major concepts and terms you will use to conduct your search—that is, your *keywords*. These, along with the research questions you identified in Chapter 11 "Writing from Research: What Will I Learn?", Section 11.2 "Steps in Developing a Research Proposal", will help you find sources using any of the following methods:

- Using the library's online catalog or card catalog
- Using periodicals indexes and databases
- Consulting a reference librarian

You probably already have some keywords in mind based on your preliminary research and writing. Another way to identify useful keywords is to visit the Library of Congress's website at <http://id.loc.gov/authorities>. This site allows you to search for a topic and see the related subject headings used by the Library of Congress, including broader terms, narrower terms, and related terms. Other libraries use these terms to classify materials. Knowing the most-used terms will help you speed up your keyword search.

Jorge used the Library of Congress site to identify general terms he could use to find resources about low-carb dieting. His search helped him identify potentially useful keywords and related topics, such as carbohydrates in human nutrition, glycemic index, and carbohydrates—metabolism. These terms helped Jorge refine his search.

Tip

Knowing the right keywords can sometimes make all the difference in conducting a successful search. If you have trouble finding sources on a topic, consult a librarian to see whether you need to modify your search terms.

Exercise 2:

Visit the Library of Congress's website at <http://id.loc.gov/authorities> and conduct searches on a few terms related to your topic.

- Review your search results and identify six to eight additional terms you might use when you conduct your research.
- Print out the search results or save the results to your research folder on your computer or portable storage device.

Using Periodicals, Indexes, and Databases

Library catalogs can help you locate book-length sources, as well as some types of nonprint holdings, such as CDs, DVDs, and audio books. To locate shorter sources, such as magazine and journal articles, you will need to use a periodical index or an online periodical database. These tools index the articles that appear in newspapers, magazines, and journals. Like catalogs, they provide publication information about an article and often allow users to access a summary or even the full text of the article.

Print indexes may be available in the periodicals section of your library. Increasingly, libraries use online databases that users can access through the library website. A single library may provide access to multiple periodical databases. These can range from general news databases to specialized databases. Table 11.2 describes some commonly used indexes and databases.

Table 11.2 Commonly Used Indexes and Databases

Resource	Format	Contents
<i>New York Times Index</i>	Print	Guide to articles published in the <i>New York Times</i>
ProQuest	Online	Database that archives content from newspapers, magazines, and dissertations
PsycLit, PsycINFO	Online	Databases that archive content from journals in psychology and psychiatry
Business Source Complete	Online	Database that archives business-related content from magazines and journals
MEDLINE, PubMed	Online	Databases that archive articles in medicine and health
EBSCOhost	Online	General database that provides access to articles on a wide variety of topics

Reading Popular and Scholarly Periodicals

When you search for periodicals, be sure to distinguish among different types. Mass-market publications, such as newspapers and popular magazines, differ from scholarly publications in their accessibility, audience, and purpose.

Newspapers and magazines are written for a broader audience than scholarly journals. Their content is usually quite accessible and easy to read. Trade magazines that target readers within a particular industry may presume the reader has background knowledge, but these publications are still reader-friendly for a broader audience. Their purpose is to inform and, often, to entertain or persuade readers as well.

Scholarly or academic journals are written for a much smaller and more expert audience. The creators of these publications assume that most of their readers are already familiar with the main topic of the journal. The target audience is also highly educated. Informing is the primary purpose of a scholarly journal. While a journal article may advance an agenda or advocate a position, the content will

still be presented in an objective style and formal tone. Entertaining readers with breezy comments and splashy graphics is not a priority.

Because of these differences, scholarly journals are more challenging to read. That doesn't mean you should avoid them. On the contrary, they can provide in-depth information unavailable elsewhere. Because knowledgeable professionals carefully review the content before publication, scholarly journals are far more reliable than much of the information available in popular media. Seek out academic journals along with other resources. Just be prepared to spend a little more time processing the information.

Writing at Work

Periodicals databases are not just for students writing research papers. They also provide a valuable service to workers in various fields. The owner of a small business might use a database such as Business Source Premiere to find articles on management, finance, or trends within a particular industry. Health care professionals might consult databases such as MedLine to research a particular disease or medication. Regardless of what career path you plan to pursue, periodicals databases can be a useful tool for researching specific topics and identifying periodicals that will help you keep up with the latest news in your industry.

Consulting a Reference Librarian

Sifting through library stacks and database search results to find the information you need can be like trying to find a needle in a haystack. If you are not sure how you should begin your search, or if it is yielding too many or too few results, you are not alone. Many students find this process challenging, although it does get easier with experience. One way to learn better search strategies is to consult a reference librarian.

Reference librarians are intimately familiar with the systems libraries use to organize and classify information. They can help you locate a particular book in the library stacks, steer you toward useful reference works, and provide tips on how to use databases and other electronic research tools. Take the time to see what resources you can find on your own, but if you encounter difficulties, ask for help. Many university librarians hold virtual office hours and are available for online chatting.

Exercise 3:

Visit your library's website or consult with a reference librarian to determine what periodicals indexes or databases would be useful for your research. Depending on your topic, you may rely on a general news index, a specialized index for a particular subject area, or both. Search the catalog for your topic and related keywords. Print out or bookmark your search results.

1. Identify at least one to two relevant periodicals, indexes, or databases.
2. Conduct a keyword search to find potentially relevant articles on your topic.
3. Save your search results. If the index you are using provides article summaries, read these to determine how useful the articles are likely to be.
4. Identify at least three to five articles to review more closely. If the full article is available online, set aside time to read it. If not, plan to visit our library within the next few days to locate the articles you need.

Tip

One way to refine your keyword search is to use Boolean operators. These operators allow you to combine keywords, find variations on a word, and otherwise expand or limit your results.

Note

Here are some of the ways you can use Boolean operators:

- Combine keywords with **and** or **+** to limit results to citations that include both keywords—for example, **diet + nutrition**.
- Combine keywords with **not** or **-** to search for the first word without the second. This can help you eliminate irrelevant results based on words that are similar to your search term. For example, searching for **obesity not childhood** locates materials on obesity but excludes materials on childhood obesity.
- Enclose a phrase in quotation marks to search for an exact phrase, such as **"morbid obesity."**
- Use parentheses to direct the order of operations in a search string. For example, since Type II diabetes is also known as adult-onset diabetes, you could search **(Type II or adult-onset) and diabetes** to limit your search results to articles on this form of the disease.
- Use a wildcard symbol such as **#**, **?**, or **\$** after a word to search for variations on a term. For instance, you might type **diabet#** to search for information on diabetes and diabetics. The specific symbol used varies with different databases.

Finding and Using Electronic Resources

With the expansion of technology and media over the past few decades, a wealth of information is available to you in electronic format. Some types of resources, such as a television documentary, may only be available electronically. Other resources—for instance, many newspapers and magazines—may be available in both print and electronic form. The following are some of the electronic sources you might consult:

- Online databases
- CD-ROMs
- Popular web search engines
- Websites maintained by businesses, universities, nonprofit organizations, or government agencies
- Newspapers, magazines, and journals published on the web
- E-books
- Audio books
- Industry blogs
- Radio and television programs and other audio and video recordings
- Online discussion groups

The techniques you use to locate print resources can also help you find electronic resources efficiently. Libraries usually include CD-ROMs, audio books, and audio and video recordings among their holdings. You can locate these materials in the catalog using a keyword search. The same Boolean operators used to refine database searches can help you filter your results in popular search engines.

Using Internet Search Engines Efficiently

When faced with the challenge of writing a research paper, some students rely on popular search engines as their first source of information. Typing a keyword or phrase into a search engine instantly pulls up links to dozens, hundreds, or even thousands of related websites—what could be easier? Unfortunately, despite its apparent convenience, this research strategy has the following drawbacks to consider:

- **Results do not always appear in order of reliability.** The first few hits that appear in search results may include sites whose content is not always reliable, such as online encyclopedias that can be edited by any user. Because websites are created by third parties, the search engine cannot tell you which sites have accurate information.
- **Results may be too numerous for you to use.** The amount of information available on the web is far greater than the amount of information housed within a particular library or database. Realistically, if your web search pulls up thousands of hits, you will not be able to visit every site—and the most useful sites may be buried deep within your search results.
- **Search engines are not connected to the results of the search.** Search engines find websites that people visit often and list the results in order of popularity. The search engine, then, is not connected to any of the results. When you cite a source found through a search engine, you do not need to cite the search engine. Only cite the source.

A general web search can provide a helpful overview of a topic and may pull up genuinely useful resources. To get the most out of a search engine, however, use strategies to make your search more efficient. Use multiple keywords and Boolean operators to limit your results. Click on the Advanced Search link on the homepage to find additional options for streamlining your search. Depending on the specific search engine you use, the following options may be available:

- Limit results to websites that have been updated within a particular time frame.
- Limit results by language or country.

- Limit results to scholarly works available online.
- Limit results by file type.
- Limit results to a particular domain type, such as .edu (school and university sites) or .gov (government sites). This is a quick way to filter out commercial sites, which can often lead to more objective results.

Use the Bookmarks or Favorites feature of your web browser to save and organize sites that look promising.

Using Other Information Sources: Interviews

With so many print and electronic media readily available, it is easy to overlook another valuable information resource: other people. Consider whether you could use a person or group as a primary source. For instance, you might interview a professor who has expertise in a particular subject, a worker within a particular industry, or a representative from a political organization. Interviews can be a great way to get firsthand information.

To get the most out of an interview, you will need to plan ahead. Contact your subject early in the research process and explain your purpose for requesting an interview. Prepare detailed questions. Open-ended questions, rather than questions with simple yes-or-no answers, are more likely to lead to an in-depth discussion. Schedule a time to meet, and be sure to obtain your subject's permission to record the interview. Take careful notes and be ready to ask follow-up questions based on what you learn.

Tip

If scheduling an in-person meeting is difficult, consider arranging a telephone interview or asking your subject to respond to your questions via e-mail. Recognize that any of these formats takes time and effort. Be prompt and courteous, avoid going over the allotted interview time, and be flexible if your subject needs to reschedule.

Evaluating Research Resources

As you gather sources, you will need to examine them with a critical eye. Smart researchers continually ask themselves two questions: "Is this source relevant to my purpose?" and "Is this source reliable?" The first question will help you avoid wasting valuable time reading sources that stray too far from your specific topic and research questions. The second question will help you find accurate, trustworthy sources.

Determining Whether a Source Is Relevant

At this point in your research process, you may have identified dozens of potential sources. It is easy for writers to get so caught up in checking out books and printing out articles that they forget to ask themselves how they will use these resources in their research. Now is a good time to get a little ruthless. Reading and taking notes takes time and energy, so you will want to focus on the most relevant sources.

To weed through your stack of books and articles, skim their contents. Read quickly with your research questions and subtopics in mind. Table 11.3 "Tips for Skimming Books and Articles" explains how to skim to get a quick sense of what topics are covered. If a book or article is not especially relevant, put it aside. You can always come back to it later if you need to.

Table 11.3 Tips for Skimming Books and Articles

Tips for Skimming Books	Tips for Skimming Articles
1. Read the dust jacket and table of contents for a broad overview of the topics covered. 2. Use the index to locate more specific topics and see how thoroughly they are covered. 3. Flip through the book and look for subtitles or key terms that correspond to your research.	1. Skim the introduction and conclusion for summary material. 2. Skim through subheadings and text features such as sidebars. 3. Look for keywords related to your topic. 4. Journal articles often begin with an abstract or summary of the contents. Read it to determine the article's relevance to your research.

Determining Whether a Source Is Reliable

All information sources are not created equal. Sources can vary greatly in terms of how carefully they are researched, written, edited, and reviewed for accuracy. Common sense will help you identify obviously questionable sources, such as tabloids that feature tales of alien abductions, or personal websites with glaring typos. Sometimes, however, a source's reliability—or lack of it—is not so obvious. For more information about source reliability, see Chapter 12 "Writing a Research Paper".

To evaluate your research sources, you will use critical thinking skills consciously and deliberately. You will consider criteria such as the type of source, its intended purpose and audience, the author's (or authors') qualifications, the publication's reputation, any indications of bias or hidden agendas, how current the source is, and the overall quality of the writing, thinking, and design.

Evaluating Types of Sources

The different types of sources you will consult are written for distinct purposes and with different audiences in mind. This accounts for other differences, such as the following:

- How thoroughly the writers cover a given topic
- How carefully the writers research and document facts
- How editors review the work
- What biases or agendas affect the content

A journal article written for an academic audience for the purpose of expanding scholarship in a given field will take an approach quite different from a magazine feature written to inform a general audience. Textbooks, hard news articles, and websites approach a subject from different angles as well. To some extent, the type of source provides clues about its overall depth and reliability. Table 11.4 ranks different source types.

Table 11.4 Source Rankings

High-Quality Sources	
These sources provide the most in-depth information. They are researched and written by subject matter experts and are carefully reviewed.	• Scholarly books and articles in scholarly journals • Trade books and magazines geared toward an educated general audience, such as <i>Smithsonian Magazine</i> or <i>Nature</i> • Government documents, such as books, reports, and web pages • Documents posted online by reputable organizations, such as universities and research institutes • Textbooks and reference books, which are usually reliable but may not cover a topic in great depth
Varied-Quality Sources	
These sources are often useful. However, they do not cover subjects in as much depth as high-quality sources, and they are not always rigorously researched and reviewed. Some, such as popular magazine articles or company brochures, may be written to market a product or a cause. Use them with caution.	• News stories and feature articles (print or online) from reputable newspapers, magazines, or organizations, such as <i>Newsweek</i> or the Public Broadcasting Service • Popular magazine articles, which may or may not be carefully researched and fact checked • Documents published by businesses and nonprofit organizations
Questionable Sources	
These sources should be avoided. They are often written primarily to attract a large readership or present the author's opinions and are not subject to careful review.	• Loosely regulated or unregulated media content, such as Internet discussion boards, blogs, free online encyclopedias, talk radio shows, television news shows with obvious political biases, personal websites, and chat rooms

Tip

Free online encyclopedias and wikis may seem like a great source of information. They usually appear among the first few results of a web search. They cover thousands of topics, and many articles use an informal, straightforward writing style. Unfortunately, these sites have no control system for researching, writing, and reviewing articles. Instead, they rely on a community of users to police themselves. At best, these sites can be a starting point for finding other, more trustworthy sources. Never use them as final sources.

Evaluating Credibility and Reputability

Even when you are using a type of source that is generally reliable, you will still need to evaluate the author's credibility and the publication itself on an individual basis. To examine the author's credibility—that is, how much you can believe of what the author has to say—examine his or her credentials. What career experience or academic study shows that the author has the expertise to write about this topic?

Keep in mind that expertise in one field is no guarantee of expertise in another, unrelated area. For instance, an author may have an advanced degree in physiology, but this credential is not a valid qualification for writing about psychology. Check credentials carefully.

Just as important as the author's credibility is the publication's overall reputability. Reputability refers to a source's standing and reputation as a respectable, reliable source of information. An established and well-known newspaper, such as the *New York Times* or the *Wall Street Journal*, is more reputable than a college newspaper put out by comparatively inexperienced students. A website that is maintained by a well-known, respected organization and regularly updated is more reputable than one created by an unknown author or group.

If you are using articles from scholarly journals, you can check databases that keep count of how many times each article has been cited in other articles. This can be a rough indication of the article's quality or, at the very least, of its influence and reputation among other scholars.

Checking for Biases and Hidden Agendas

Whenever you consult a source, always think carefully about the author's or authors' purpose in presenting the information. Few sources present facts completely objectively. In some cases, the source's content and tone are significantly influenced by biases or hidden agendas.

Bias refers to favoritism or prejudice toward a particular person or group. For instance, an author may be biased against a certain political party and present information in a way that subtly—or not so subtly—makes that organization look bad. Bias can lead an author to present facts selectively, edit quotations to misrepresent someone's words, and distort information.

Hidden agendas are goals that are not immediately obvious but influence how an author presents the facts. For instance, an article about the role of beef in a healthy diet would be questionable if it were written by a representative of the beef industry—or by the president of an animal-rights organization. In both cases, the author would likely have a hidden agenda.

As Jorge conducted his research, he read several research studies in which scientists found significant benefits to following a low-carbohydrate diet. He also noticed that many studies were sponsored by a foundation associated with the author of a popular series of low-carbohydrate diet books. Jorge read these studies with a critical eye, knowing that a hidden agenda might be shaping the researchers' conclusions.

Using Current Sources

Be sure to seek out sources that are current, or up to date. Depending on the topic, sources may become outdated relatively soon after publication, or they may remain useful for years. For instance, online social networking sites have evolved rapidly over the past few years. An article published in 2002 about this topic will not provide current information. On the other hand, a research paper on elementary education practices might refer to studies published decades ago by influential child psychologists.

When using websites for research, check to see when the site was last updated. Many sites publish this information on the homepage, and some, such as news sites, are updated daily or weekly. Many nonfunctioning links are a sign that a website is not regularly updated. Do not be afraid to ask your professor for suggestions if you find that many of your most relevant sources are not especially reliable—or that the most reliable sources are not relevant.

Evaluating Overall Quality by Asking Questions

When you evaluate a source, you will consider the criteria previously discussed as well as your overall impressions of its quality. Read carefully, and notice how well the author presents and supports his or her statements. Stay actively engaged—do not simply accept an author's words as truth. Ask questions to determine each source's value. Checklist 11.1 lists ten questions to ask yourself as a critical reader.

Note

Source Evaluation Checklist 11.1:

- Is the type of source appropriate for my purpose? Is it a high-quality source or one that needs to be looked at more critically?
- Can I establish that the author is credible and the publication is reputable?
- Does the author support ideas with specific facts and details that are carefully documented? Is the source of the author's information clear? (When you use secondary sources, look for sources that are not too removed from primary research.)
- Does the source include any factual errors or instances of faulty logic?
- Does the author leave out any information that I would expect to see in a discussion of this topic?
- Do the author's conclusions logically follow from the evidence that is presented? Can I see how the author got from one point to another?
- Is the writing clear and organized, and is it free from errors, clichés, and empty buzzwords? Is the tone objective, balanced, and reasonable? (Be on the lookout for extreme, emotionally charged language.)
- Are there any obvious biases or agendas? Based on what I know about the author, are there likely to be any hidden agendas?
- Are graphics informative, useful, and easy to understand? Are websites organized, easy to navigate, and free of clutter like flashing ads and unnecessary sound effects?
- Is the source contradicted by information found in other sources? (If so, it is possible that your sources are presenting similar information but taking different perspectives, which requires you to think carefully about which sources you find more convincing and why. Be suspicious, however, of any source that presents facts that you cannot confirm elsewhere.)

Writing at Work

The critical thinking skills you use to evaluate research sources as a student are equally valuable when you conduct research on the job. If you follow certain periodicals or websites, you have probably identified publications that consistently provide reliable information. Reading blogs and online discussion groups is a great way to identify new trends and hot topics in a particular field, but these sources should not be used for substantial research.

Exercise 4:

Use a search engine to conduct a web search on your topic. Refer to the tips provided earlier to help you streamline your search. Evaluate your search results critically based on the criteria you have learned. Identify and bookmark one or more websites that are reliable, reputable, and likely to be useful in your research.

Managing Source Information

As you determine which sources you will rely on most, it is important to establish a system for keeping track of your sources and taking notes. There are several ways to go about it, and no one system is necessarily superior. What matters is that you keep materials in order; record bibliographical information you will need later; and take detailed, organized notes.

Keeping Track of Your Sources

Think ahead to a moment a few weeks from now, when you've written your research paper and are almost ready to submit it for a grade. There is just one task left—writing your list of sources.

As you begin typing your list, you realize you need to include the publication information for a book you cited frequently. Unfortunately, you already returned it to the library several days ago. You do not remember the URLs for some of the websites you used or the dates you accessed them—information that also must be included in your bibliography. With a sinking feeling, you realize that finding this information and preparing your bibliography will require hours of work.

This stressful scenario can be avoided. Taking time to organize source information now will ensure that you are not scrambling to find it at the last minute. Throughout your research, record bibliographical information for each source as soon as you begin using it. You may use pen-and-paper methods, such as a notebook or note cards, or maintain an electronic list. (If you prefer the latter option, many office software packages include separate programs for recording bibliographic information.)

Table 11.5 "Details for Commonly Used Source Types" shows the specific details you should record for commonly used source types. Use these details to develop a working bibliography—a preliminary list of sources that you will later use to develop the references section of your paper. You may wish to record information using the formatting system of the American Psychological Association (APA) or the Modern Language Association (MLA), which will save a step later on. (For more information on APA and MLA formatting, see Chapter 13 "APA and MLA Documentation and Formatting".)

Table 11.5 Details for Commonly Used Source Types

Source Type	Necessary Information
Book	Author(s), title and subtitle, publisher, city of publication, year of publication
Essay or article published in a book	Include all the information you would for any other book. Additionally, record the essay's or article's title, author(s), the pages on which it appears, and the name of the book's editor(s).
Periodical	Author(s), article title, publication title, date of publication, volume and issue number, and page numbers
Online source	Author(s) (if available), article or document title, organization that sponsors the site, database name (if applicable), date of publication, date you accessed the site, and URL
Interview	Name of person interviewed, method of communication, date of interview

Your research may involve less common types of sources not listed in Table 11.5 "Details for Commonly Used Source Types". For additional information on citing different sources, see Chapter 13 "APA and MLA Documentation and Formatting".

Exercise 5:

Create a working bibliography using the format that is most convenient for you. List at least five sources you plan to use. Continue to add sources to your working bibliography throughout the research process.

Tip

To make your working bibliography even more complete, you may wish to record additional details, such as a book's call number or contact information for a person you interviewed. That way, if you need to locate a source again, you have all the information you need right at your fingertips. You may also wish to assign each source a code number to use when taking notes (1, 2, 3, or a similar system).

Taking Notes Efficiently

Good researchers stay focused and organized as they gather information from sources. Before you begin taking notes, take a moment to step back and think about your goal as a researcher—to find information that will help you answer your research question. When you write your paper, you will present your conclusions about the topic supported by research. That goal will determine what information you record and how you organize it.

Writers sometimes get caught up in taking extensive notes, so much so that they lose sight of how their notes relate to the questions and ideas they started out with. Remember that you do not need to write down every detail from your reading. Focus on finding and recording details that will help you answer your research questions. The following strategies will help you take notes efficiently.

Use Headings to Organize Ideas

Whether you use old-fashioned index cards or organize your notes using word-processing software, record just one major point from each source at a time, and use a heading to summarize the information covered. Keep all your notes in one file, digital or otherwise. Doing so will help you identify connections among different pieces of information. It will also help you make connections between your notes and the research questions and subtopics you identified earlier.

Know When to Summarize, Paraphrase, or Directly Quote a Source

Your notes will fall under three categories—summary notes, paraphrased information, and direct quotations from your sources. Effective researchers make choices about which type of notes is most appropriate for their purpose.

- Summary notes sum up the main ideas in a source in a few sentences or a short paragraph. A summary is considerably shorter than the original text and captures only the major ideas. Use summary notes when you do not need to record specific details but you intend to refer to broad concepts the author discusses.
- Paraphrased notes restate a fact or idea from a source using your own words and sentence structure.
- Direct quotations use the exact wording used by the original source and enclose the quoted material in quotation marks. It is a good strategy to copy direct quotations when an author expresses an idea in an especially lively or memorable way. However, do not rely exclusively on direct quotations in your note taking.

Most of your notes should be paraphrased from the original source. Paraphrasing as you take notes is usually a better strategy than copying direct quotations, because it forces you to think through the information in your source and understand it well enough to restate it. In short, it helps you stay engaged with the material instead of simply copying and pasting. Synthesizing will help you later when you begin planning and drafting your paper. (For detailed guidelines on summarizing, paraphrasing, and quoting, see Chapter 11 "Writing from Research: What Will I Learn?", Section 11.6 "Writing from Research: End-of-Chapter Exercises".)

Maintain Complete, Accurate Notes

Regardless of the format used, any notes you take should include enough information to help you organize ideas and locate them instantly in the original text if you need to review them. Make sure your notes include the following elements:

- Heading summing up the main topic covered
- Author's name, a source code, or an abbreviated source title
- Page number
- Full URL of any pages buried deep in a website

Throughout the process of taking notes, be scrupulous about making sure you have correctly attributed each idea to its source. Always include source information so you know exactly which ideas came from which sources. Use quotation marks to set off any words for phrases taken directly from the original text. If you add your own responses and ideas, make sure they are distinct from ideas you quoted or paraphrased.

Finally, make sure your notes accurately reflect the content of the original text. Make sure quoted material is copied verbatim. If you omit words from a quotation, use ellipses to show the omission and make sure the omission does not change the author's meaning. Paraphrase ideas carefully, and check your paraphrased notes against the original text to make sure that you have restated the author's ideas accurately in your own words.

Use a System That Works for You

There are several formats you can use to take notes. No technique is necessarily better than the others—it is more important to choose a format you are comfortable using. Choosing the format that works best for you will ensure your notes are organized, complete, and accurate. Consider implementing one of these formats when you begin taking notes:

- **Use index cards.** This traditional format involves writing each note on a separate index card. It takes more time than copying and pasting into an electronic document, which encourages you to be selective in choosing which ideas to record. Recording notes on separate cards makes it easy to later organize your notes according to major topics. Some writers color-code their cards to make them still more organized.
- **Use note-taking software.** Word-processing and office software packages often include different types of note-taking software. Although you may need to set aside some time to learn the software, this method combines the speed of typing with the same degree of organization associated with handwritten note cards.
- **Maintain a research notebook.** Instead of using index cards or electronic note cards, you may wish to keep a notebook or electronic folder, allotting a few pages (or one file) for each of your sources. This method makes it easy to create a separate column or section of the document where you add your responses to the information you encounter in your research.
- **Annotate your sources.** This method involves making handwritten notes in the margins of sources that you have printed or photocopied. If using electronic sources, you can make comments within the source document. For example, you might add comment boxes to a PDF version of an article. This method works best for experienced researchers who have already thought a great deal about the topic because it can be difficult to organize your notes later when starting your draft.

Choose one of the methods from the list to use for taking notes. Continue gathering sources and taking notes. In the next section, you will learn strategies for organizing and synthesizing the information you have found.

Conclusion

- A writer's use of primary and secondary sources is determined by the topic and purpose of the research. Sources used may include print sources, such as books and journals; electronic sources, such as websites and articles retrieved from databases; and human sources of information, such as interviews.
- Strategies that help writers locate sources efficiently include conducting effective keyword searches, understanding how to use online catalogs and databases, using strategies to narrow web search results, and consulting reference librarians.
- Writers evaluate sources based on how relevant they are to the research question and how reliable their content is.
- Skimming sources can help writers determine their relevance efficiently.
- Writers evaluate a source's reliability by asking questions about the type of source (including its audience and purpose); the author's credibility, the publication's reputability, the source's currency, and the overall quality of the writing, research, logic, and design in the source.
- In their notes, effective writers record organized, complete, accurate information. This includes bibliographic information about each source as well as summarized, paraphrased, or quoted information from the source.

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CHAPTER OVERVIEW

3: Chapter 3

3.1: Information Literacy

Evaluate sources

Information formats

Research tools

Search strategies

The information timeline

The research cycle

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CHAPTER OVERVIEW

3.1: Information Literacy

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Evaluate sources

Information sources vary in quality, and before you use a source in your academic assignments or work projects, you must evaluate them for quality. You want your own work to be of high quality, credible, and accurate, and you can only achieve that by having sources possess those same qualities.

Watch [this video on evaluating sources](#) for an overview of what credibility is, why it's important, and some of the criteria to look for when evaluating a source.



Video credit: [Evaluating Sources for Credibility](#) by NCSU Libraries, CC: BY-NC-SA 3.0 US

There are five basic criteria for evaluating information. While it may seem like a lot to think about at first, after a little practice, you will find that you can evaluate sources quickly.

Here are the five basic criteria, with key questions and indicators to help you evaluate your source:

- Authority
 - **Key Question:** Is the person, organization, or institution responsible for the intellectual content of the information knowledgeable in that subject?
 - **Indicators of authority:** formal academic degrees, years of professional experience, active and substantial involvement in a particular area
- Accuracy
 - **Key Question:** How free from error is this piece of information?
 - **Indicators of accuracy:** correct and verifiable citations, information is verifiable in other sources from different authors/organizations, author is authority on subject
- Objectivity
 - **Key Question:** How objective is this piece of information?
 - **Indicators of objectivity:** multiple points of view are acknowledged and discussed logically and clearly, statements are supported with documentation from a variety of reliable sources, purpose is clearly stated
- Currency
 - **Key Question:** When was the item of information published or produced?
 - **Indicators of currency:** publication date, assignment restrictions (e.g., you can only use articles from the last 5 years), your topic and how quickly information changes in your field (e.g., technology or health topics will require very recent information to reflect rapidly changing areas of expertise)
- Audience
 - **Key Question:** Who is this information written for or this product developed for?
 - **Indicators of audience:** language, style, tone, bibliographies

When evaluating and selecting sources for an assignment or work project, compare your sources to one another in light of your topic. Imagine, for example, you are writing a paper about bicycle commuting. You have three sources about bicycle safety. One is written for children; one is for adult recreational bicyclers; and one is for traffic engineers. Your topic is specifically about building urban and suburban infrastructure to encourage bicycling, so the source written for traffic engineers is clearly more appropriate for your topic than the other two. Even if the other two are high-quality sources, they are not the most relevant sources for your specific topic.

Until you have practiced evaluating many sources, it can be a little difficult to find the indicators, especially in web sources.

Table 6. Locations where you might find indicators of quality in different types of sources

If you are looking for indications of...	In books see the...	In journals see the...	In websites see the...
Authority	Title page, Forward, Preface, Afterward, Dust Jackets, Bibliography	Periodical covers, Editorial Staff, Letters to the Editor, Abstract, Bibliography	URL, About Us, Publications, Appearance
Accuracy	Title page, Forward, Preface, Afterward, Periodical covers, Dust Jackets, Text, Bibliography	Periodical covers, Text, Bibliography	URL, About Us, Home page, Awards, Text
Objectivity	Forward, Preface, Afterward, Text, Bibliography	Abstracts, Text, Bibliography, Editorials, Letters to the Editor	About Us, Site Map, Text, Disclaimers, Membership/Registration
Currency	Title page, copyright page, Bibliography	Title page, Bibliography, Abstracts	Home page, Copyright, What's New
Audience	Forward, Preface, Afterward	Letters to the editor, Editorial, Appearance	Home page, About, Mission, Disclaimer, Members only

When you have located and evaluated information for your paper, report, or project, you will use it to complete your work. A later chapter of this text will cover how to use it correctly and ethically.

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Information formats

Traditionally, information has been organized in different formats, usually because of the time it takes to gather and publish the information. For example, the purpose of news reporting is to inform the public about the basic facts of an event. This information needs to be disseminated quickly, so it is published daily in print, online, on broadcast television, and radio media.

Today, publication of information in traditional formats continues as well as in constantly evolving formats on the Internet. These new information formats can include electronic journals, e-books, news websites, blogs, Twitter, Facebook, and other social media sites. The coexistence of all of these information formats is messy and chaotic, and the process for finding relevant information is not always clear.

One way to make some sense out of the current information universe is to understand thoroughly traditional information formats. You can then understand the concepts inherent in the information formats found online. There are some direct correlations such as books and journal articles, but there are also some newer formats like tweets that didn't exist until recently.

Primary and secondary information sources

Primary sources allow researchers to get as close as possible to original ideas, events, and empirical research as possible. Such sources may include creative works, first hand or contemporary accounts of events, and the publication of the results of empirical observations or research.

Examples of primary sources are interviews, letters, emails, Tweets, Facebook posts, photographs, speeches, newspaper or magazine articles written at the time of an event, works of literature, lab notes, field research, and published scientific research.

Secondary sources analyze, review, or summarize information in primary resources or other secondary resources. Even sources presenting facts or descriptions about events are secondary unless they are based on direct participation or observation. Secondary sources analyze the primary sources.

Examples of secondary sources are journal articles, books, literature reviews, literary criticism, meta-analyses of scientific studies, documentaries, biographies, and textbooks.

Sometimes the line between primary and secondary sources blurs. For example, although newspapers and news websites contain primary source material, they also contain secondary source material. For example, an article published on November 6, 2012 about the results of the US presidential election would be a primary source, because it was written on the day of the event. However, an article published in the same paper two weeks later analyzing how the successful candidate raised money for his campaign would be a secondary source.

Table 1. Examples of primary and secondary sources for technical writing

Primary	Secondary
Interview with subject matter expert	Documentary on an issue or problem
Survey data	News article about scientific study
Published scientific study	Literature review on a research topic

Popular, professional, and scholarly information

At some point in your college career, you will be asked to find peer-reviewed resources on your research topic. Your professor may explain that these appear in scholarly journals. You may wonder what makes a scholarly journal article different from an article in a magazine, like *National Geographic* or *Sports Illustrated*.

Popular magazines like *People*, *Sports Illustrated*, and *Rolling Stone* can be good sources for articles on recent events or pop-culture topics, while magazines like *Harper's*, *Scientific American*, and *The New Republic* will offer more in-depth articles on a wider range of subjects. The audience for these publications are readers who, although not experts, are knowledgeable about the issues presented.

Professional journals, also called trade journals, address an audience of professionals in a specific discipline or field. They report news and trends in a field, but not original research. They may also provide product or service reviews, job listings, and advertisements.

Scholarly journals provide articles of interest to experts or researchers in a discipline. An editorial board of respected scholars in a discipline (peers of the authors) reviews all articles submitted to a journal. They decide if the article provides a noteworthy contribution to the field and should be published. Scholarly journals contain few or no advertisements. Scholarly journal articles will include references of works cited and may also have footnotes or endnotes, all of which rarely appear in popular or professional publications.

Peer review is a widely accepted indicator of quality scholarship in a given discipline or field. Peer-reviewed (or refereed) journals are scholarly journals that only publish articles that have passed through this review process.

You can better understand peer-review by watching the video “[Peer Review in 3 Minutes.](#)”

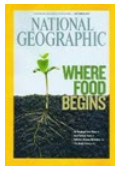
PEER REVIEW



IN 3 MINUTES

Video credit: [Peer Review in 3 Minutes](#) by NCSU Libraries, [CC: BY-NC-SA 3.0 US](#)

Table 2. Differences among magazines, professional journals, and scholarly journals

Example	 Magazine	 Professional journal	 Academic journal
Audience	General public	People employed in a field	Researchers, scholars, experts
Bibliography	No	Sometimes	Yes
Article length/depth Structure	- Overview - Current events - General interest articles	- Articles will be of interest to those working in that field - Purpose will be to offer advice and tips to those in the trade	- Long and narrowly focused - Specialized vocabulary - Article structure will usually contain abstract, literature review, methodology, results, conclusion, references
Review policy	Magazine editor	Magazine editor and possibly a board	Editorial board/ scholars in the field Peer-reviewed
Author	Journalist or specialist	Someone working in the field	Researcher/expert in the field
Appearance	- Glossy - Many graphics - Many advertisements	- Glossy - Advertising specific to that trade	- Lengthy articles - Often includes charts, graphs, statistics - Little or no advertising

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Research tools

Databases

Very often, you will want articles on your topic, and the easiest way to find articles is to use a research database – a specialized search engine for finding articles and other types of content. For example, if your topic is residential solar power, you can use a research database to look in thousands of journal titles at once and find the latest scientific and technical research articles – articles that don’t always show up in your Google searches. Depending on the database you are using, you might also find videos, images, diagrams, or e-books on your topic, too. Most databases require subscriptions for access; check with your college, public, or corporate library to see what databases they subscribe to for you to use. Whether they are subscription-based or free, research databases contain records of journal articles, documents, book chapters, and other resources and are not tied to the physical items available at any one library. Many databases provide the full-text of articles and can be searched by keyword, subject, author, or title. A few databases provide just the citations for articles, but they usually also provide tools for you to find the full text in another database or request it through interlibrary loan. Databases are made up of records, and records contain fields, as explained below:

- Records: A record describes one information item (e.g., journal article, book chapter, image, etc.).
- Fields: These are part of the record, and they contain descriptions of specific elements of the information item such as the title, author, publication date, and subject.

Another aspect of databases to know about is controlled vocabulary. Look for the label subject term, subject heading or descriptor. Regardless of label, this field contains controlled vocabulary, which are designated terms or phrases for describing concepts. It’s important because it pulls together all of the items in that database about one topic. For example, imagine you are searching for information on community colleges. Different authors may call community colleges by different names: junior colleges, two-year colleges, or technical colleges. If the controlled vocabulary term in the database you are searching is “community colleges,” then your one search will pull up all results, regardless of what term the author uses.

Databases may seem intimidating at first, but you likely use databases in everyday life. For example, do you store contact information in your phone? If so, you create a record for everyone for whom you want to store information. In each of those records, you enter descriptions into fields: first and last name, phone number, email address, and physical address. If you wanted to organize your contacts, you might put them into groups of “work,” “family,” and “friends.” That would be your controlled vocabulary for your own database of contacts.

The image below shows a detailed record for a journal article from a common research database, Academic Search Premier. The fields and controlled vocabulary are labeled.

Figure 2. Fields and controlled vocabulary



The screenshot shows the EBSCOhost search interface. At the top, there's a navigation bar with links like 'New Search', 'Publications', 'Subject Terms', 'Cited References', 'More', 'Sign In', 'Folder', and 'Preferences'. Below this, the search bar contains 'technical writing'. To the right of the search bar are buttons for 'Search' and 'Clear'. Below the search bar, there are three rows of 'AND' connectors and 'Select a Field (optional)' dropdowns. Below these are links for 'Basic Search', 'Advanced Search', and 'Search History'. The search results are displayed in a list format. The first result is titled 'Ten Simple (Empirical) Rules for Writing Science.' and is from 'PLOS Computational Biology'. The authors listed are Weinberger, Cody J., Evans, James A., and Allesina, Stefano. The source is 'PLOS Computational Biology, Apr2015, Vol. 11 Issue 4, p1-6. 6p. 2 Diagrams.' The document type is 'Editorial'. The subject terms are: *TECHNICAL writing, *AUTHORSHIP, *LANGUAGE & languages, *TECHNOLOGY -- Terminology, *ADVERBS (Grammar), and *PRESENT tense (Grammar). On the left side of the search results, there are links for 'Detailed Record' and 'PDF Full Text (290KB)'. A box labeled 'Fields' is drawn around the left side of the search results, and a box labeled 'Controlled Vocabulary' is drawn around the subject terms.

General

Databases that contain resources for many subject areas are referred to as general or multidisciplinary databases. This means that they are good starting points for research because they allow you to search a large number of sources from a wide variety of disciplines. Content types in general databases often include a mix of professional publications, scholarly journals, newspapers, magazines, books, and multimedia.

Common general databases in academic settings include

- Academic Search Premier
- Academic OneFile
- JSTOR

Specialized

Databases devoted to a single subject are known as subject-specific or specialized databases. Often, they search a smaller number of journals or a specific type of content. Specialized databases can be very powerful search tools after you have selected a narrow research topic or if you already have a great deal of expertise in a particular area. They will help you find information you would not find in a general database. If you are not sure which specialized databases are available for your topic, check your library's website for subject guides, or ask a librarian.

Specialized databases may also focus on offering one specific content type like streaming films, music, images, statistics, or data sets.

Table 4. Examples of specialized databases and their subject focus

Database	Subject Focus
PsycInfo	Psychology
BioOne	Life Sciences
MEDLINE	Medicine/health
ARTstor	Fine arts images

Library catalogs

A library catalog is a database that contains all of the items located in a library as well as all of the items to which the library offers access, either in physical or online format. It allows you to search for items by title, author, subject, and keyword. Like research databases, library catalogs use controlled vocabulary to allow for powerful searching using specific terms or phrases.

If you locate a physical item in a catalog, you will need the call number to find the item in the library. A call number is like a street address for a book; it tells you exactly where the book is on the shelf. Most academic libraries will use the Library of Congress classification system, and the call numbers start with letters, followed by a mix of numbers and letters. In addition to helping you find a book, call numbers group items about a given topic together in a physical place. The image below shows an example of the location of a call number in a library catalog.

Figure 3. A call number in an online library catalog display



Along with physical items, most libraries also provide access to scholarly e-books, streaming films, and e-journals via their catalogs. For students and faculty, these resources are usually available from any computer, meaning you do not have to go to the library and retrieve items from the shelf.

Consortia and interlibrary loan

In the course of your research you will almost certainly find yourself in a situation where you have a citation for a journal article or book that your institution's library doesn't have. There is a wealth of knowledge contained in the resources of academic and public libraries throughout the United States. Single libraries cannot hope to collect all of the resources available on a topic. Fortunately, libraries are happy to share their resources and they do this through consortia membership and interlibrary loan (ILL). Library consortia are groups of libraries that have special agreements with one another to loan materials to one another's users. Both academic and public libraries belong to consortia.

Central Oregon Community College (COCC) belongs to a consortium of 37 academic libraries in the Pacific Northwest called the Orbis Cascade Alliance, which sponsors a shared lending program called Summit. When you search the COCC Library Catalog, your results will include items from other Summit institutions, which you can request and have delivered to the COCC campus of your choice.

Interlibrary loan (ILL) allows you to borrow books, articles and other information resources regardless of where they are located. If you find an article when searching a database that COCC doesn't have full text access to, you can request it through the interlibrary loan program. If you cannot get a book or DVD from COCC or another Summit library, you could also request it through interlibrary loan. Interlibrary loan services are available at both academic and public libraries.

Government information

Another important source of information is the government. Official United States government websites end in .gov and provide a wealth of credible information, including statistics, technical reports, economic data, scientific and medical research, and, of course, legislative information. Unlike research databases, government information is typically freely available without a subscription.

[USA.gov](https://www.usa.gov/) is a search engine for government information and is a good place to begin your search, though specialized search tools are also available for many topics. State governments also have their own websites and search tools that you might find helpful if your topic has a state-specific angle. If you get lost in searching government information, ask a librarian for help. They usually have special training and knowledge in navigating government information.

Table 5. Examples of specialized government sources and their subject focus

Specialized Government Source	Subject Focus
PubMed Central	Medicine/health
ERIC	Education

Experts

People are a valuable, though often overlooked, source. This might be particularly appropriate if you are working on an emerging topic or a topic with local connections.

For personal interviews, there are specific steps you can take to obtain better results. Do some background work on the topic before contacting the person you hope to interview. The more familiarity you have with your topic and its terminology, the easier it will be to ask focused questions. Focused questions are important for effective research. Asking general questions because you think the specifics might be too detailed rarely leads to the best information. Acknowledge the time and effort someone is taking to answer your questions, but also realize that people who are passionate about subjects enjoy sharing what they know. Take the opportunity to ask experts about additional resources they would recommend.

For a successful, productive interview, review this list of [Interview Tips](#) before conducting your interview.

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Search strategies

Now that you know more about the research tools available to you, it's time to consider how to construct searches that will allow you to find the most relevant, useful results as efficiently as possible.

Develop effective keywords

The single most important search strategy is to choose effective search terms. This may seem obvious, but it is too often overlooked, with deleterious consequences. When deciding what terms to use in a search, break down your topic into its main concepts. Do not enter an entire sentence or a full question. The best thing to do is to use the key concepts involved with your topic. In addition, think of synonyms or related terms for each concept. If you do this, you will have more flexibility when searching in case your first search term does not produce any or enough results. This may sound strange, since if you are looking for information using a Web search engine, you usually get too many results. Databases, however, contain fewer items than the entire web, and having alternative search terms may lead you to useful sources. Even in a search engine like Google, having terms you can combine thoughtfully will yield better results.

Figure 4. Keyword brainstorming for the topic of violence in high schools

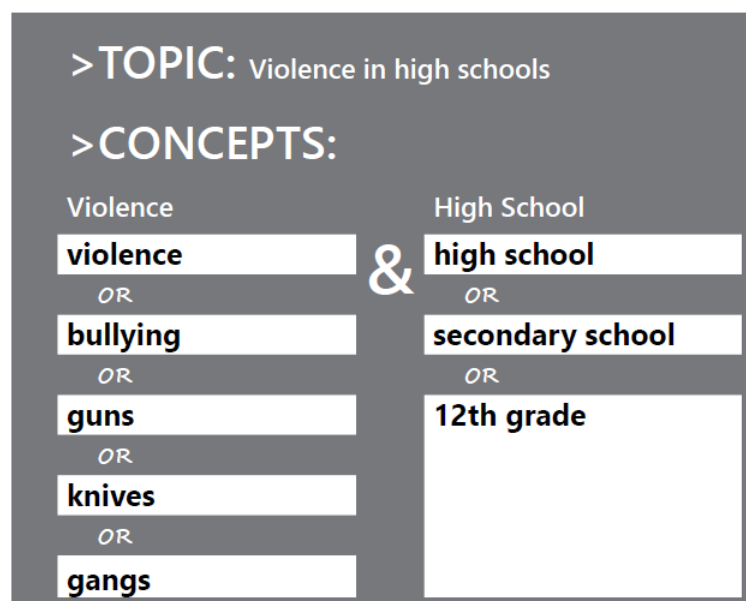


Image credit: [The Information Literacy User's Guide](#) edited by Greg Bobish and Trudi Jacobson, CC: BY-NC-SA 3.0 US

Advanced search techniques

Once you have identified the concepts you want to search and have carefully chosen your keywords, think about how you will enter them into the search box of your selected search tool. Try the techniques below in both research databases and web search engines.

Boolean operators

Boolean operators are a search technique that will help you

- focus your search, particularly when your topic contains multiple search terms
- connect various pieces of information to find exactly what you are looking for

There are three Boolean operators: AND, OR, and NOT. You capitalize Boolean operators to distinguish them from the words and, or, and not (words which most search engines ignore). It is also important to note that you should spell out AND rather than substituting commas, ampersands, or plus signs. Usually you should spell out NOT as well, except in Google, where you must use the minus sign (-).

AND

Use AND in a search to

- narrow your results
- tell the database that ALL search terms must be present in the resulting records

Example: cloning AND humans AND ethics

The purple triangle where all circles intersect in the middle of the Venn diagram below represents the result set for this search. It is a small set created by a combination of all three search words.

Figure 5. Example of search terms connected by AND

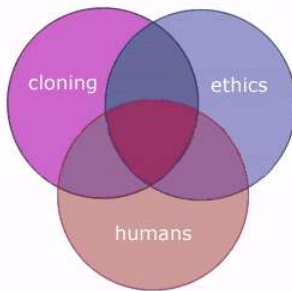


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OR

Use OR in a search to accomplish the following:

- connect two or more similar concepts (synonyms)
- broaden your results, telling the database that any one of your search terms can be present in the resulting records

Example: (cloning OR genetics OR reproduction)

All three circles represent the result set for this search. It is a big set because the OR operator includes all of those search terms.

Figure 6. Example of search terms connected by OR

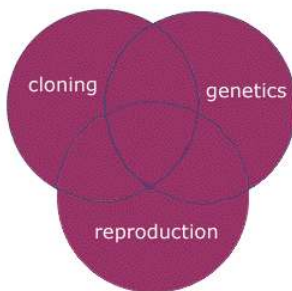


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NOT

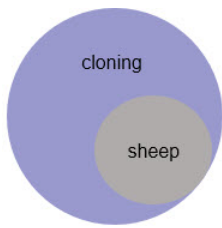
Use NOT in a search to

- exclude words from your search
- narrow your search, telling the database to ignore concepts that may be implied by your search terms

Example: cloning NOT sheep

The purple part of the circle below represents your results for this search, because you've used NOT to exclude a subset of results that are about sheep (represented by the gray circle).

Figure 7. Example of search using NOT



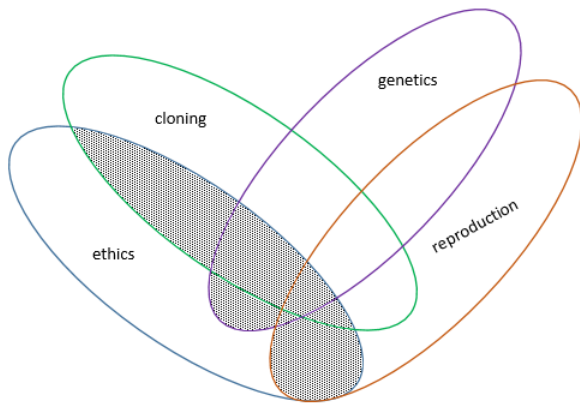
Combining operators

You can combine the different Boolean operators into one search. The important thing to know when combining operators is to use parentheses around the terms connected with OR. This ensures that the database interprets your search query correctly.

Example: ethics AND (cloning OR genetics OR reproduction)

The shaded area of the diagram below represents the results set for the above search.

Figure 8. Example of complex Boolean search that connects terms with AND and OR



Phrases

Web and research databases usually treat your search terms as separate words, meaning they look for each word appearing in a document, regardless of its location around the other words in your search term. Sometimes you may want to system to instead search for a specific phrase (a set of words that collectively describe your topic).

To do this, put the phrase in quotation marks, as in “community college.”

With phrase searches, you will typically get fewer results than searching for the words individually, which makes it an effective way to focus your search.

Truncation

Truncation, also called stemming, is a technique that allows you to search for multiple variations of a root word at once.

Most databases have a truncation symbol. The * is the most commonly used symbol, but !, ?, and # are also used. If you are not sure, check the help files.

To use truncation, enter the root of your word and end it with the truncation symbol.

Example: genetic* searches for genetic, genetics, genetically

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The information timeline

Another difference in popular, professional, and scholarly sources lies in when information appears in these types of sources. Information about an event or issue appears in publications according to a predictable pattern known as the information timeline. Familiarity with the information timeline can help you best plan your research topics and where to search for information. For example, it typically takes several months to years for information about an event or issue to appear in scholarly publications. If you choose a topic that is very recent, you may have to rely more heavily on news media, popular magazines, and primary sources (such as interviews you conduct) for your research.

Table 3. The information timeline and typical sources.

Time:	Day of event	Days later	Weeks later	Months later	Year(s) later
Sources	Television, radio, web	Newspapers, TV, radio, web	Popular and mass market magazines	Professional and scholarly journals	Scholarly journals, books, conference proceedings Reference sources such as encyclopedias
Type of information	General: who, what, where (usually not why)	Varies, some articles include analysis, statistics, photographs, editorials, opinions	Still in reporting stage, general, editorial, opinions, statistics, photographs Usually no bibliography at this stage	Research results, detailed and theoretical discussion Bibliography available at this stage	In-depth coverage of a topic, edited compilations of scholarly articles relating to a topic General overview giving factual information Bibliography available
Locating tools	Web search tools, social networks	Web search tools, newspaper and periodical databases	Web search tools, newspaper and periodical databases	General and subject-specific databases	Library catalog, general and subject-specific databases Library reference collection

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The research cycle

Although information publication follows a linear timeline, the research process itself is not linear. For example, you might start by trying to read scholarly articles, only to discover that you lack the necessary background knowledge to use a scholarly article effectively. To increase your background information, you might consult an encyclopedia or a book on your topic. Or, you may encounter a statement in a newspaper editorial that inspires you to consult the scholarly literature to see if research supports the statement.

The important thing to remember is that you will probably start your research at different points and move around among resource types depending on the type of information you need.

Figure 1. The research cycle

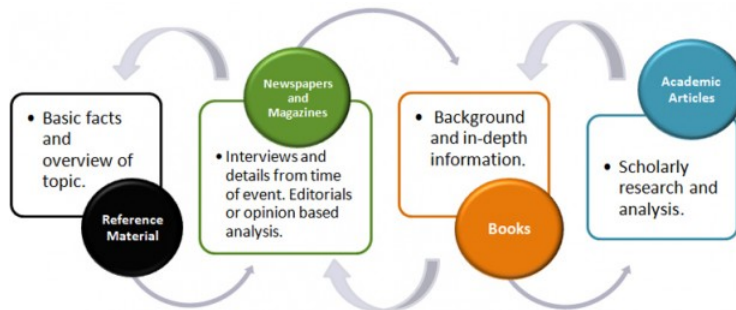


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